

Sri Lanka Institute of Information Technology



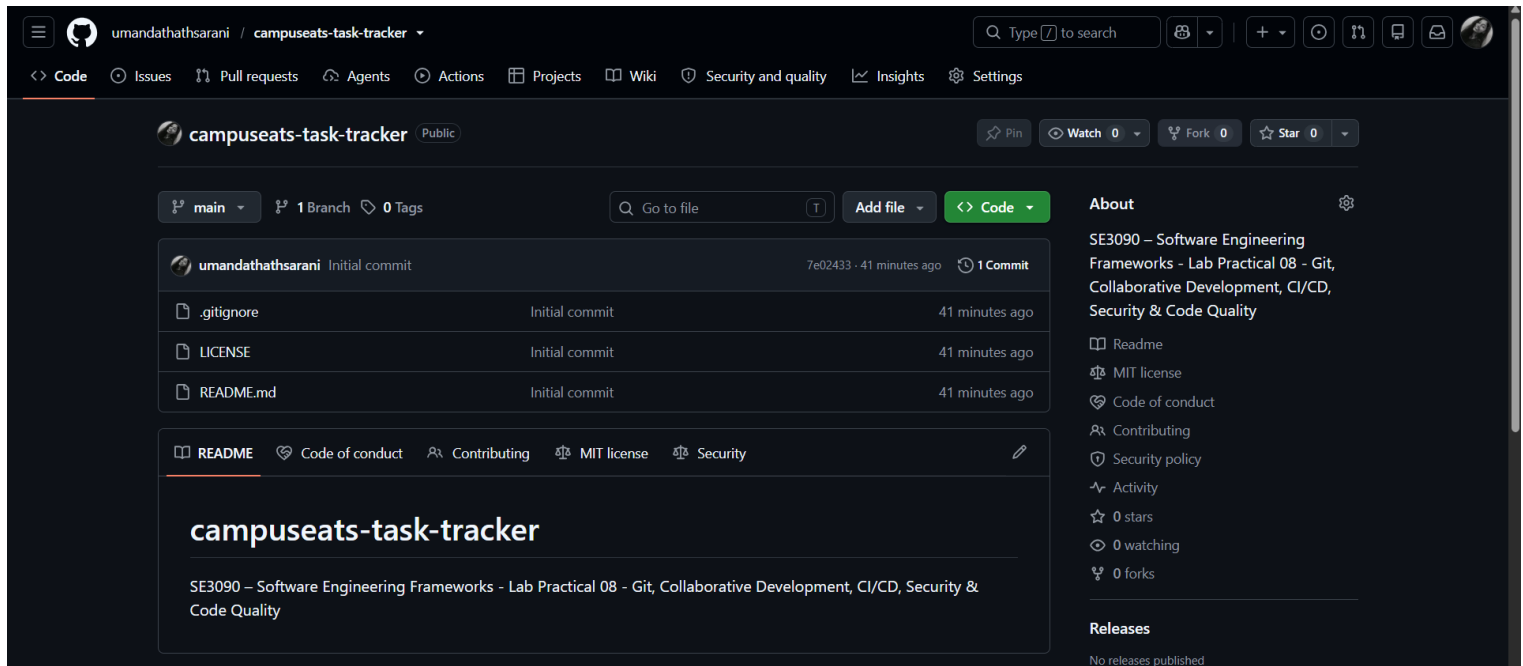
SE3090 – Software Engineering Frameworks
Year 3, Semester 1- 2026

Practical Answer Submission

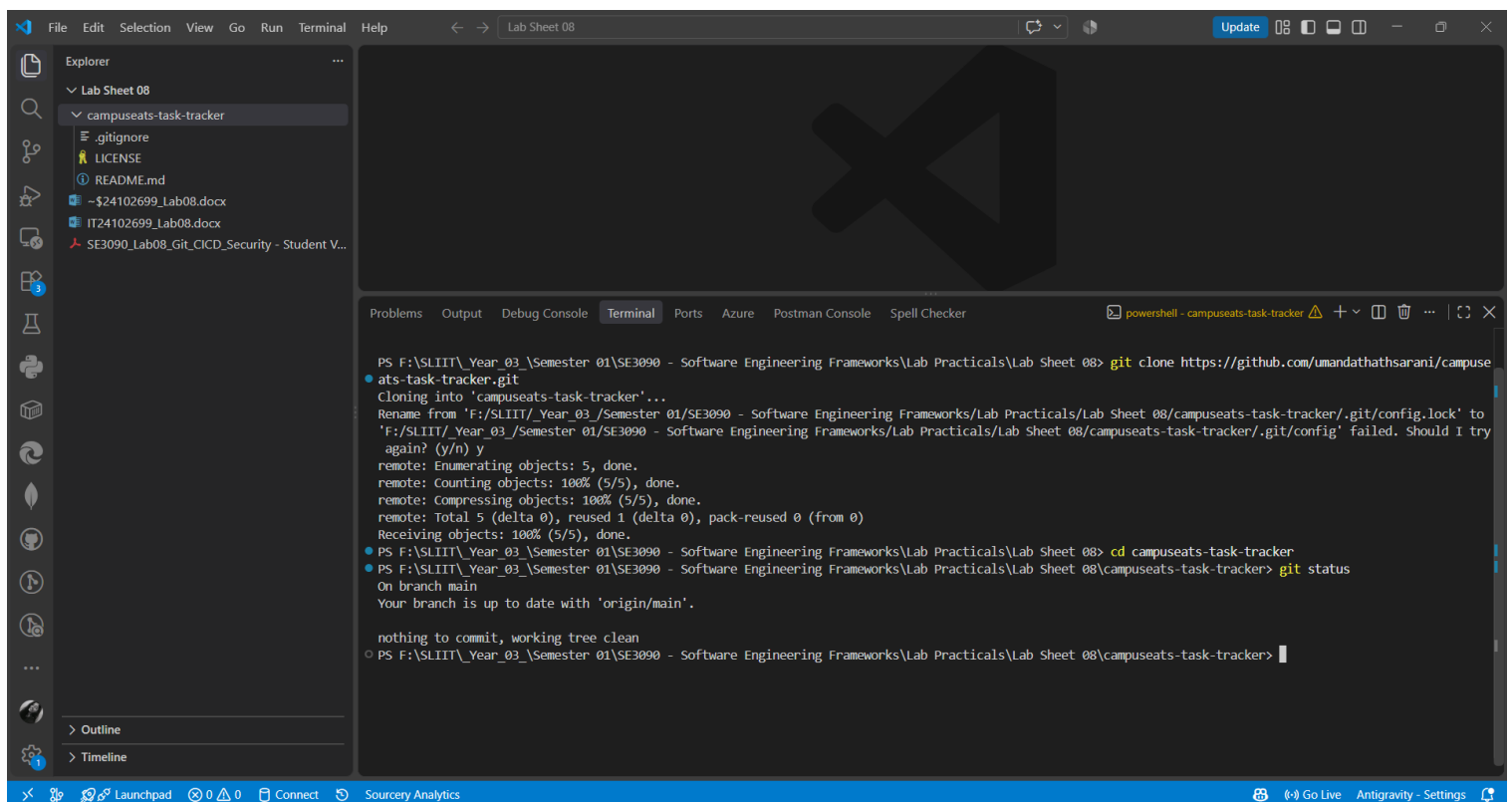
Practical Sheet No: 08

Student ID	IT24102699
Student Name	Mummullage B.U.T
Campus/ Center name	Malabe
Specialization	AI
Batch No	Y3.S1.WD.AI.01.01

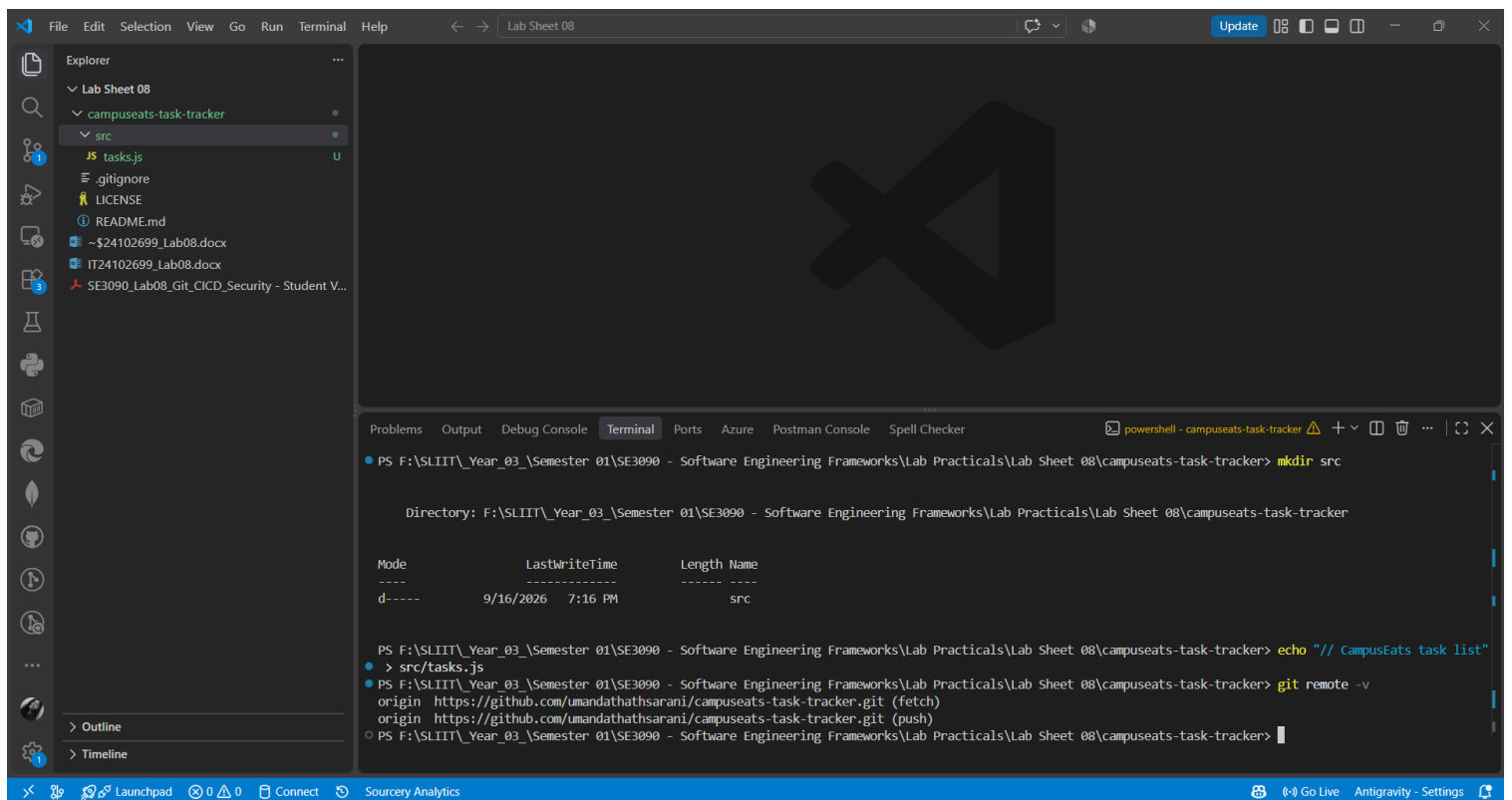
Task 01 — GitHub repository setup + clone Repo cloned locally



1. Clone the repository and move into the folder:



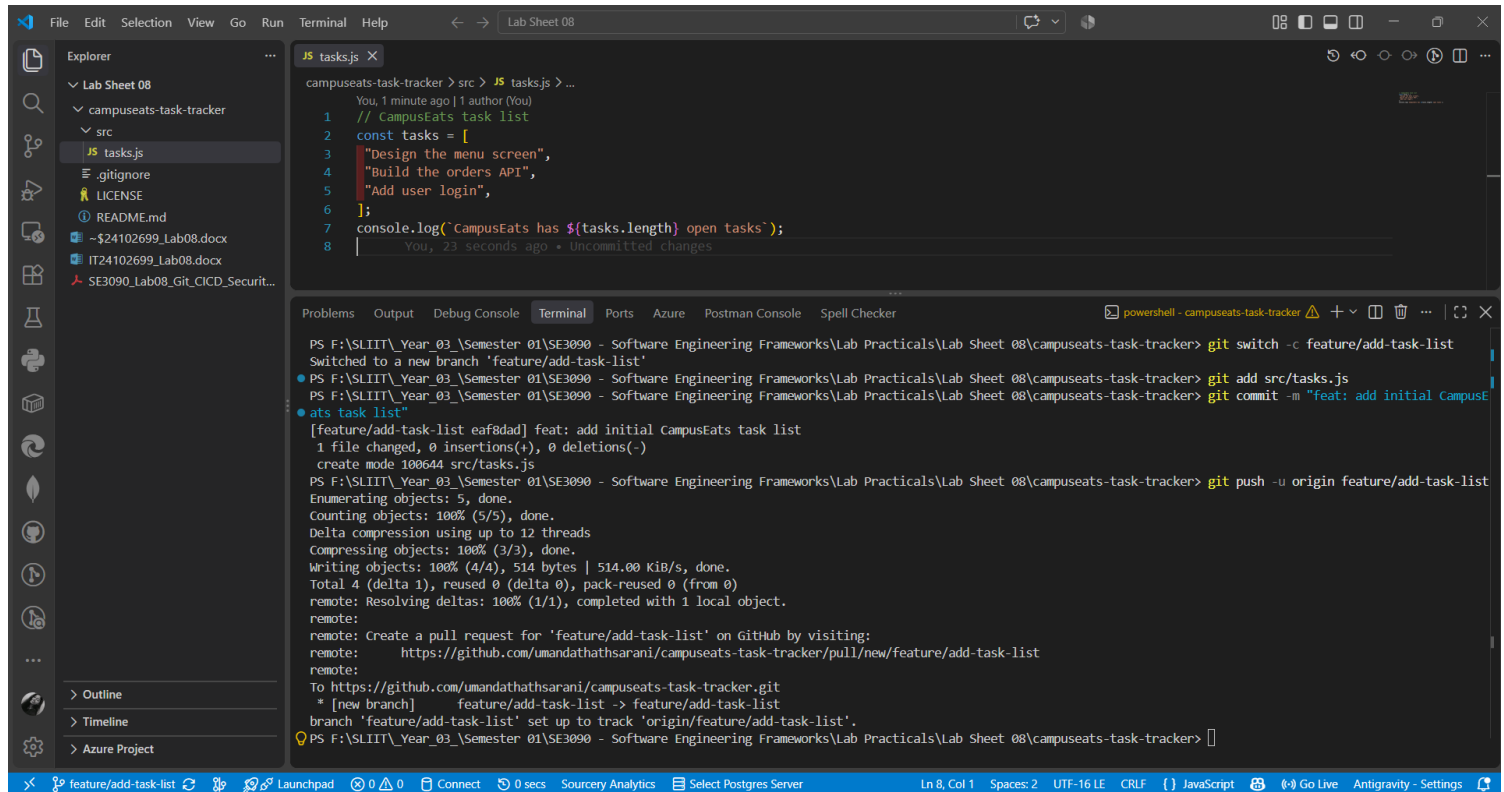
2. Create the source file and check the remote connection:



Repository Link:

<https://github.com/umandathathsarani/campuseats-task-tracker>

Task 02 — Branching and commit workflow Feature branch pushed



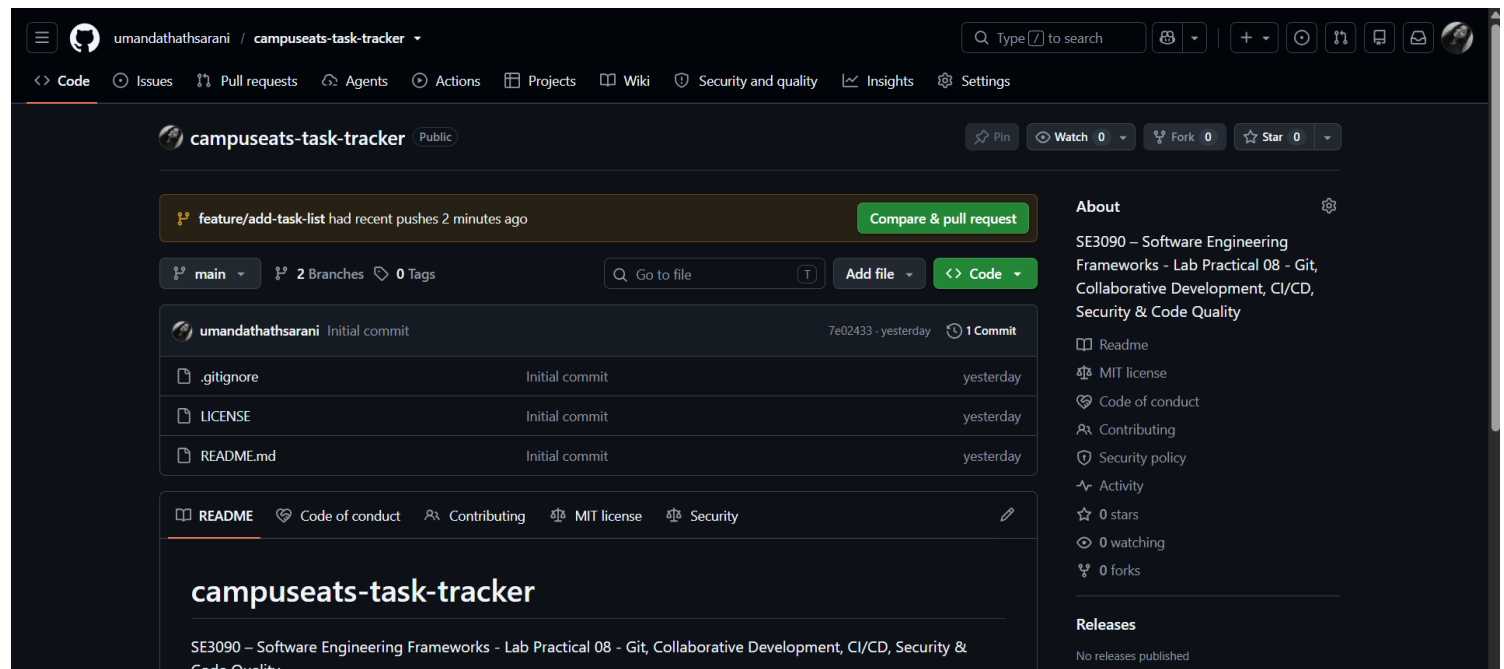
The screenshot shows the Visual Studio Code editor with the file explorer on the left displaying the project structure. The main editor shows the `tasks.js` file with the following content:

```
1 // CampusEats task list
2 const tasks = [
3   "Design the menu screen",
4   "Build the orders API",
5   "Add user login",
6 ];
7 console.log(`CampusEats has ${tasks.length} open tasks`);
8
```

The terminal at the bottom shows the execution of the following commands:

```
git switch -c feature/add-task-list
git add src/tasks.js
git commit -m "feat: add initial CampusEats task list"
git push -u origin feature/add-task-list
```

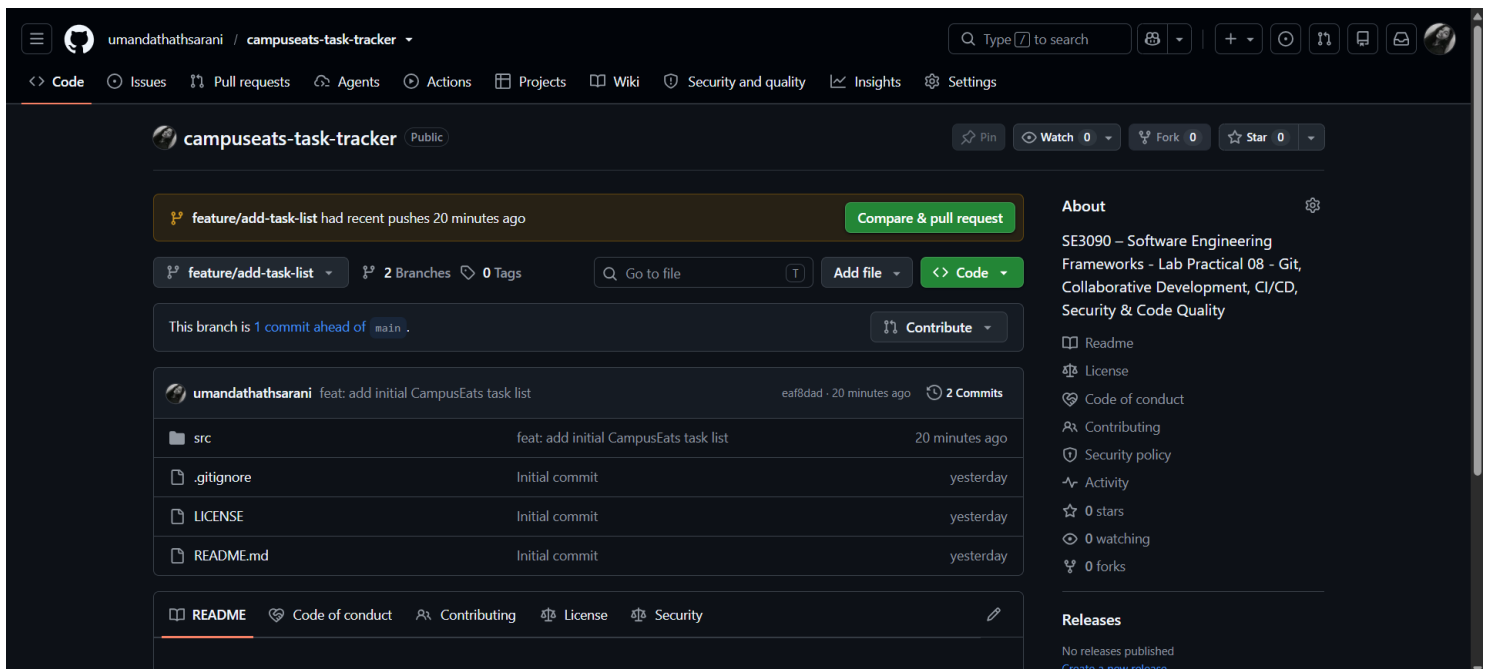
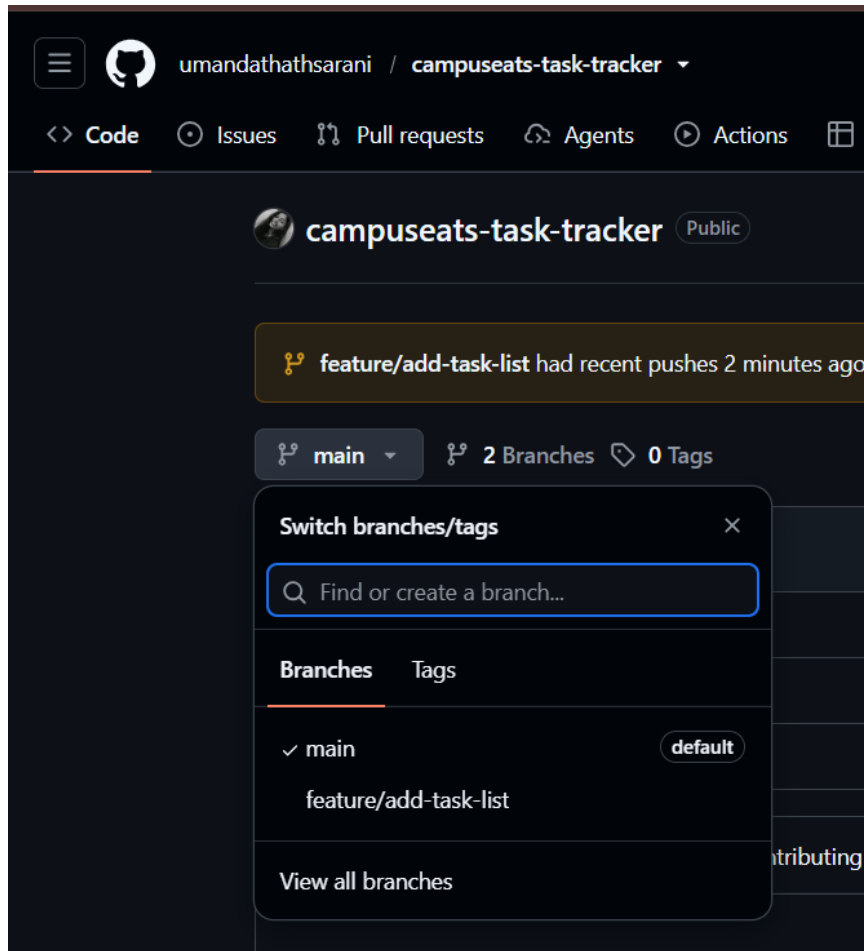
The terminal output shows the successful execution of these commands, including the creation of the new branch, staging the file, committing it, and pushing it to the remote repository.

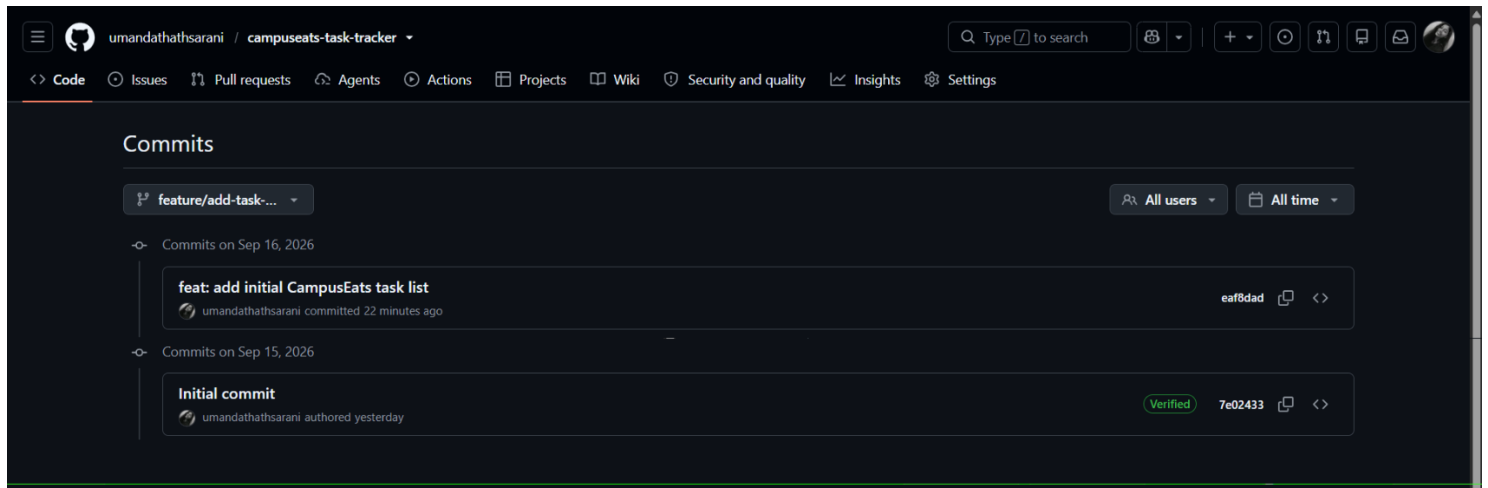


The screenshot shows the GitHub repository page for `campuseats-task-tracker`. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is `main`, and there are 2 branches in total. The repository contains the following files:

- `.gitignore` (Initial commit, yesterday)
- `LICENSE` (Initial commit, yesterday)
- `README.md` (Initial commit, yesterday)

The repository description is: "SE3090 – Software Engineering Frameworks - Lab Practical 08 - Git, Collaborative Development, CI/CD, Security & Code Quality". The repository also has a README, Code of conduct, Contributing, MIT license, and Security policy.





For this project, we are following the **GitHub Flow** branching strategy. In this workflow, the main branch is considered the single source of truth and must always remain in a stable, deployable state. Instead of committing directly to main, developers create short-lived, descriptively named feature branches (such as feature/add-task-list) for any new features, bug fixes, or updates. Once the work on a feature branch is complete, a Pull Request (PR) is opened to propose the changes. The PR serves as a dedicated space for peer code review, discussion, and automated checks. Only after the code has been reviewed, approved, and has passed all automated tests is the feature branch merged back into main. This strategy ensures high code quality, clean collaboration, and prevents broken code from reaching production.

Task 03 — Pull request and code review PR merged to main

umandathasarani / campuseats-task-tracker

Search

Type to search

<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base: main

←

compare: feature/add-task-list

✓ Able to merge. These branches can be automatically merged.

Add a title *

feat: add initial CampusEats task list

Add a description

Write

Preview

H B I

This PR adds the initial tasks.js file containing the CampusEats task list

Markdown is supported

Paste, drop, or click to add files

Create pull request

▼

Remember, contributions to this repository should follow its [contributing guidelines](#), [security policy](#), and [code of conduct](#).

1 commit

1 file changed

1 contributor

Commits on Sep 16, 2026

feat: add initial CampusEats task list

umandathasarani

committed 33 minutes ago

Showing 1 changed file with 0 additions and 0 deletions.

Split Unified

▼ BIN +360 Bytes src/tasks.js

Binary file not shown.

Reviewers

No reviews

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

Contributing

Code of conduct

Security policy

Support

GitHub Community Guidelines

© 2026 GitHub, Inc.

Terms

Privacy

Security

Status

Community

Docs

Contact

Manage cookies

Do not share my personal information

umandathasarani / campuseats-task-tracker

Type to search

<> Code

Issues

Pull requests 1

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

feat: add initial CampusEats task list #1

Ready to mergeCode

Open

umandathasarani wants to merge 1 commit into main from feature/add-task-list

Conversation 0

Commits 1

Checks 0

Files changed 1

umandathasarani commented now

Owner

This PR adds the initial tasks.js file containing the CampusEats task list

Feat: add initial CampusEats task list

eaf8dad

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? Convert to draft

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Conversation 1

Commits 1

Checks 0

Files changed 1

All commits

Filter files...

src

tasks.js

src/tasks.js

Binary file not shown.

0 / 1 viewed

Submit review

Finish your review

Write

Preview

Looks good!

Paste, drop, or click to add files

Comment

Submit general feedback without explicit approval.

Approve

Pull request authors can't approve their own pull requests.

Request changes

Pull request authors can't request changes on their own pull requests.

Cancel

Submit review

umandathasarani / campuseats-task-tracker

Type to search

<> Code

Issues

Pull requests 1

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

feat: add initial CampusEats task list #1

Ready to mergeCode

Open

umandathasarani wants to merge 1 commit into main from feature/add-task-list

Conversation 1

Commits 1

Checks 0

Files changed 1

umandathasarani commented 9 minutes ago

Owner

This PR adds the initial tasks.js file containing the CampusEats task list

Feat: add initial CampusEats task list

eaf8dad

umandathasarani commented 1 minute ago

Owner

Author

View reviewed changes

Looks good!

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

umandathathsarani commented 3 minutes ago

View reviewed changes

umandathathsarani left a comment

Owner Author ...

Looks good!

Commit message

Merge pull request #1 from umandathathsarani/feature/add-task-list

Extended description

feat: add initial CampusEats task list

This commit will be authored by umathathsarani2003@gmail.com.

Confirm merge

Cancel

Still in progress? [Convert to draft](#)

umandathathsarani / campuseats-task-tracker

Q. Type to search

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

feat: add initial CampusEats task list #1

Code

Merged umandathathsarani merged 1 commit into main from feature/add-task-list now

Conversation Commits Checks Files changed

umandathathsarani commented 12 minutes ago

Owner ...

This PR adds the initial tasks.js file containing the CampusEats task list

feat: add initial CampusEats task list

ear8dad

umandathathsarani commented 4 minutes ago

View reviewed changes

umandathathsarani left a comment

Owner Author ...

Looks good!

umandathathsarani merged commit a17bcbf into main now

Revert

Pull request successfully merged and closed

You're all set — the feature/add-task-list branch can be safely deleted.

Delete branch

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one — assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Unsubscribe

Conversation 1

Commits 1

Checks 0

Files changed 1



umandathasarani commented 19 minutes ago

Owner ...

This PR adds the initial tasks.js file containing the CampusEats task list





feat: add initial CampusEats task list

eaf8dad



umandathasarani commented 11 minutes ago

Owner Author ...

Looks good!





umandathasarani merged commit a17bcbf into main 7 minutes ago

Revert



Pull request successfully merged and closed

You're all set — the [feature/add-task-list](#) branch can be safely deleted.

Delete branch

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

 Unsubscribe

Problems

Output

Debug Console

Terminal

Ports

Azure

Postman Console

Spell Checker

powershell - campuseats-task-tracker

git switch main

Switched to branch 'main'

Your branch is up to date with 'origin/main'.

git pull origin main

remote: Enumerating objects: 1, done.

remote: Counting objects: 100% (1/1), done.

remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

Unpacking objects: 100% (1/1), 926 bytes | 2.00 KiB/s, done.

From https://github.com/umandathasarani/campuseats-task-tracker

* branch main -> FETCH_HEAD

7e02433..a17bcbf main -> origin/main

Updating 7e02433..a17bcbf

Fast-forward

src/tasks.js | Bin 0 -> 360 bytes

1 file changed, 0 insertions(+), 0 deletions(-)

create mode 100644 src/tasks.js

main

Launchpad

0 secs

Sourcery Analytics

Select Postgres Server

Ln 8, Col 1

Spaces: 2

UTF-16 LE

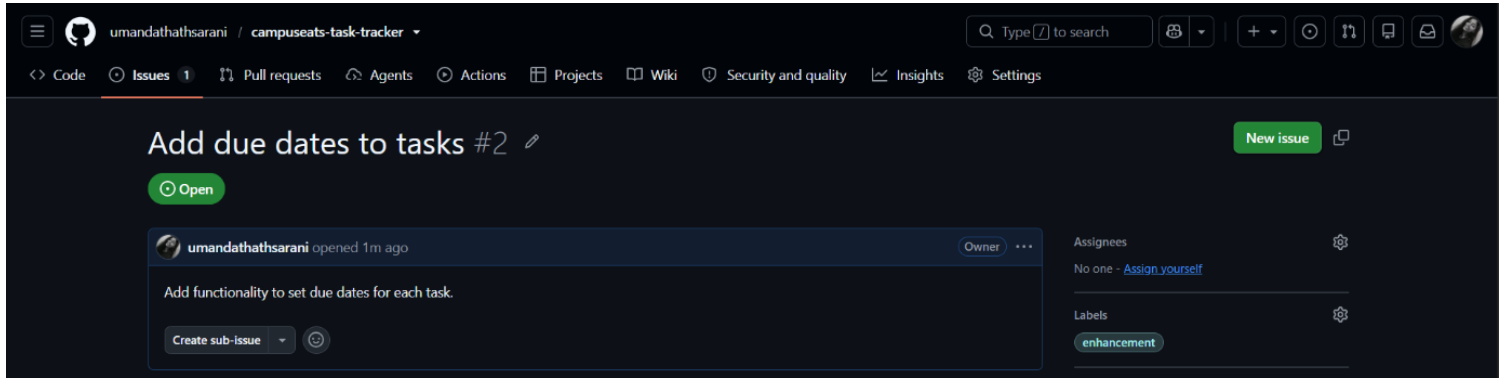
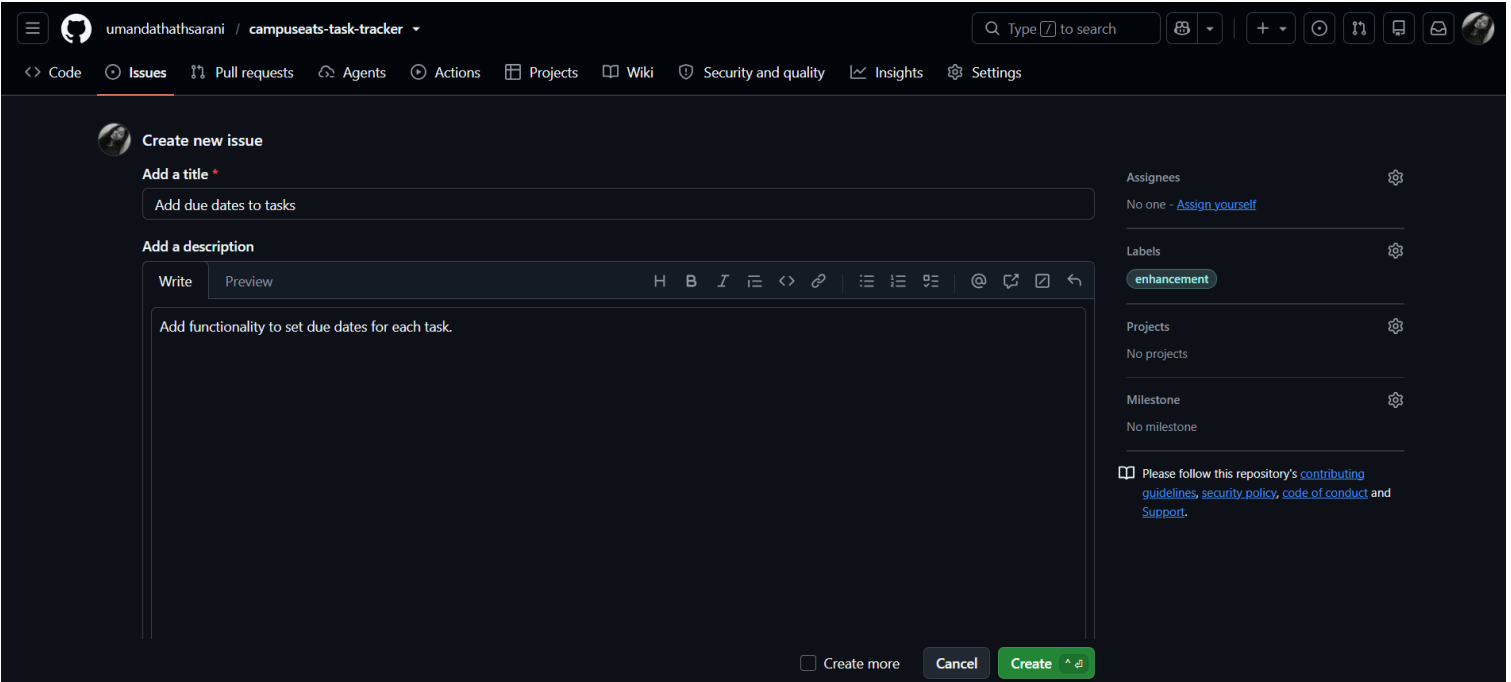
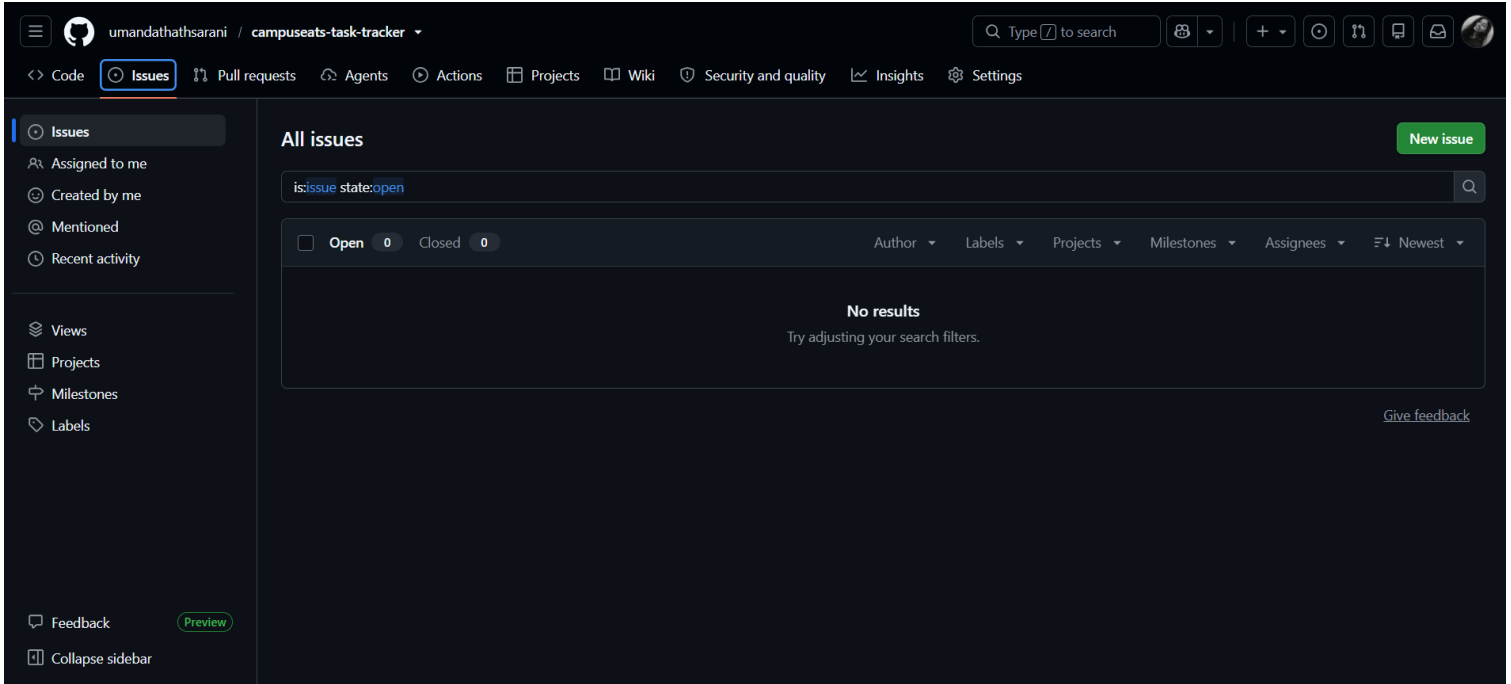
CRLF

JavaScript

Go Live

Antigravity - Settings

Task 04 — GitHub Issues and collaboration Issues created



umandathathsarani / campuseats-task-tracker

Type to search

<> Code

Issues

1

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

Create new issue

Add a title

Fix typo in README

Add a description

WritePreview

HBI~~U~~<>LinkList~~U~~ImageLink

Correct the spelling mistakes in the README file.

Assignees

No one - [Assign yourself](#)

Labels

bug

Projects

No projects

Milestone

No milestone

Please follow this repository's [contributing guidelines](#), [security policy](#), [code of conduct](#) and [Support](#).

Create more

Cancel

Create

☰

umandathasrani / campuseats-task-tracker

🔍 Type to search

🔒

+

🔄

🔗

📁

📧

🌐

<> Code

🕒 Issues 2

🔗 Pull requests

👤 Agents

🕒 Actions

📁 Projects

📖 Wiki

🔒 Security and quality

📈 Insights

⚙️ Settings

Fix typo in README #3

📝

New issue

📋

🕒 Open

🌐 umandathasrani

opened 1m ago

Owner

...

Correct the spelling mistakes in the README file.

Create sub-issue

🕒

Assignees

No one - [Assign yourself](#)

Labels

bug

umandathasarani / campuseats-task-tracker

Q Type [f] to search

<> Code Issues 2 Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Create new issue

Add a title *

Add CI workflow

Add a description

Write Preview

H B I ≡ <> 🔗 ⋮ ≡ ≡ @ 📎 ↻ ↶

Set up automated checks using GitHub Actions.

Assignees

No one - [Assign yourself](#)

Labels

enhancement

Projects

No projects

Milestone

No milestone

Please follow this repository's [contributing guidelines](#), [security policy](#), [code of conduct](#) and [Support](#).

☐ Create more

Cancel

Create ^ ⌨

☰

umandathasrani / campuseats-task-tracker

Q Type 17 to search









<> Code

 **Issues** 3

 Pull requests

 Agents

 Actions

 Projects

 Wiki

 Security and quality

 Insights

 Settings

Add CI workflow #4

 Open

 **umandathasrani** opened 1m ago

Owner

...

Set up automated checks using GitHub Actions.

Create sub-issue



  **umandathasrani** added enhancement 1m ago

Assignees

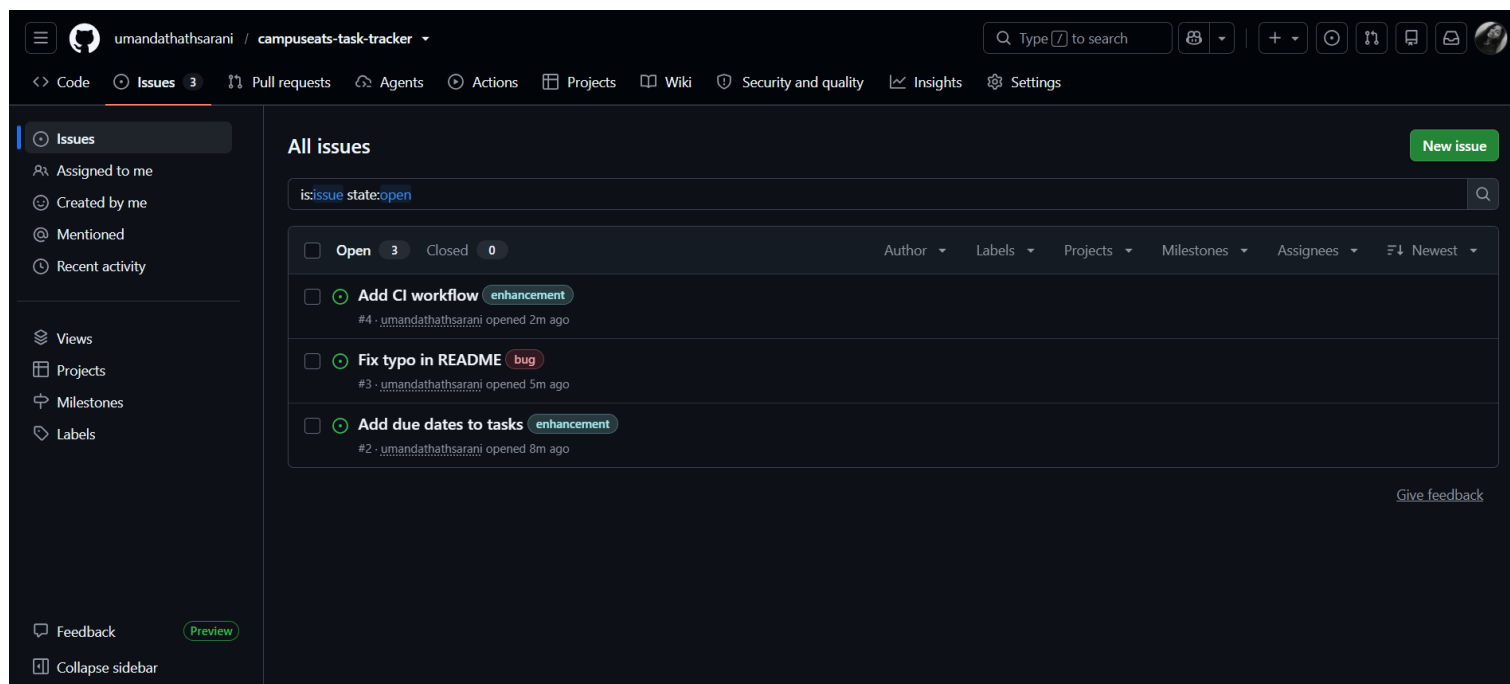
No one - [Assign yourself](#)

Labels

enhancement

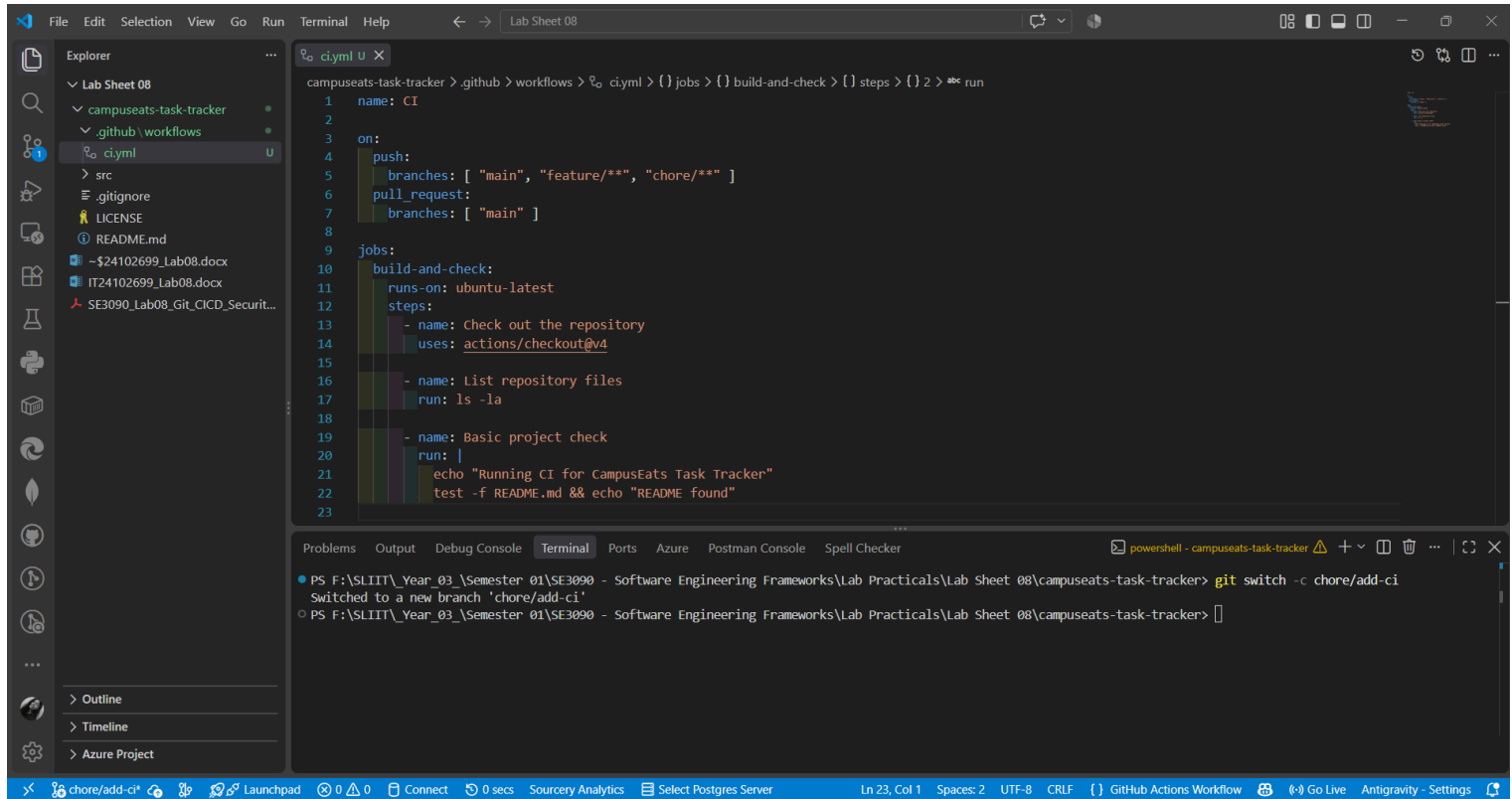
Projects

No projects



GitHub provides a helpful feature to link code changes directly to Issues using special keywords. By including a phrase like Closes #1 or Fixes #2 in a commit message or a Pull Request description, you create an automatic link between your work and the tracked issue. When that commit or PR is successfully merged into the main branch, GitHub will automatically close the linked issue for you. This keeps the project's task board organized and ensures issues are closed exactly when the fix goes live, without needing manual updates.

Task 05 — CI pipeline with GitHub Actions Green workflow run

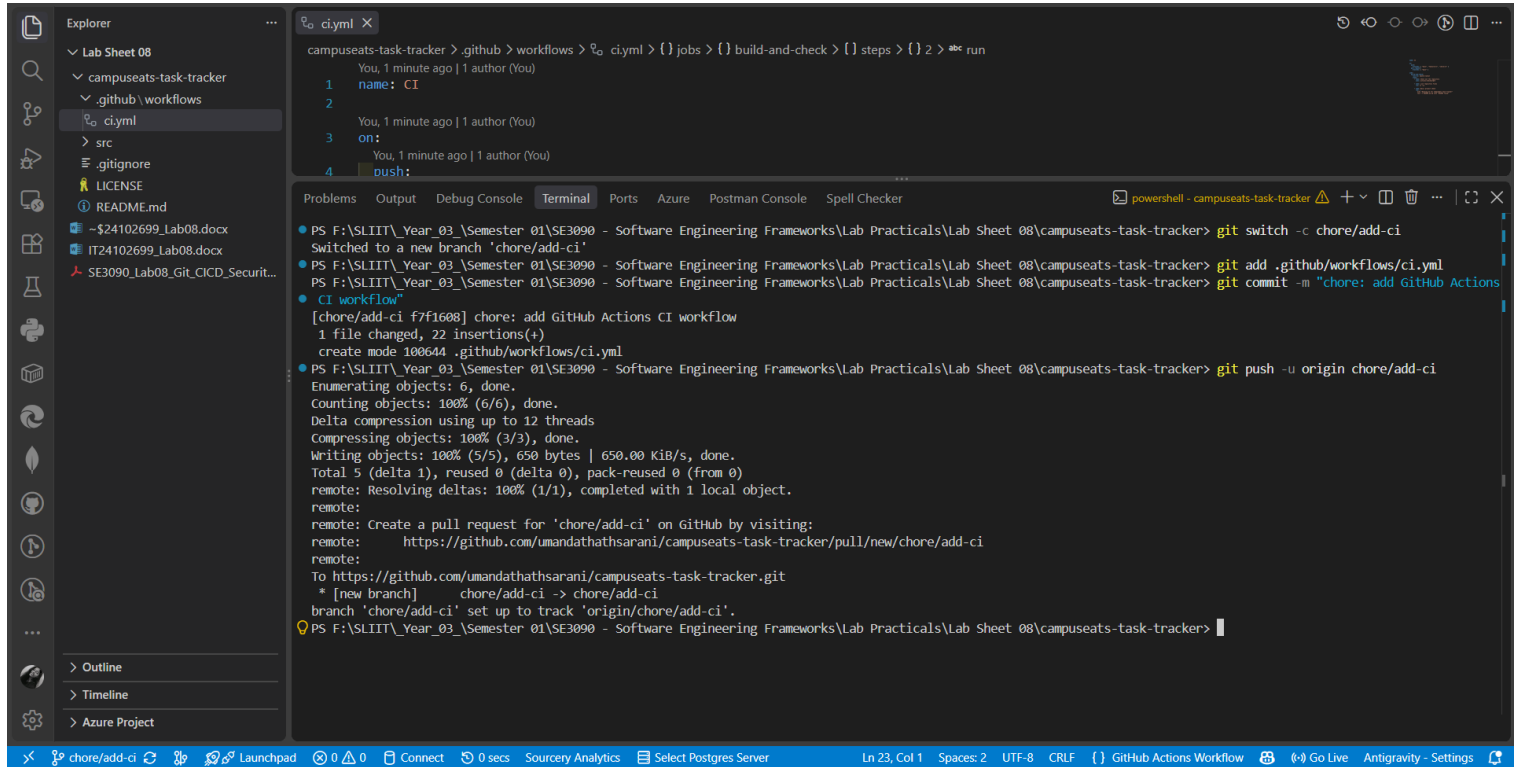


```
campuseats-task-tracker > .github > workflows > ci.yml > {} jobs > {} build-and-check > {} steps > {} 2 > abc run

1  name: CI
2
3  on:
4    push:
5      branches: [ "main", "feature/**", "chore/**" ]
6    pull_request:
7      branches: [ "main" ]
8
9  jobs:
10   build-and-check:
11     runs-on: ubuntu-latest
12     steps:
13       - name: Check out the repository
14         uses: actions/checkout@v4
15
16       - name: List repository files
17         run: ls -la
18
19       - name: Basic project check
20         run: |
21           echo "Running CI for CampusEats Task Tracker"
22           test -f README.md && echo "README found"
23
```

Problems Output Debug Console Terminal Ports Azure Postman Console Spell Checker

- PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> git switch -c chore/add-ci
- Switched to a new branch 'chore/add-ci'
- PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> |



```
campuseats-task-tracker > .github > workflows > ci.yml > {} jobs > {} build-and-check > {} steps > {} 2 > abc run

1  name: CI
2
3  on:
4    push:
5      branches: [ "main", "feature/**", "chore/**" ]
6    pull_request:
7      branches: [ "main" ]
8
9  jobs:
10   build-and-check:
11     runs-on: ubuntu-latest
12     steps:
13       - name: Check out the repository
14         uses: actions/checkout@v4
15
16       - name: List repository files
17         run: ls -la
18
19       - name: Basic project check
20         run: |
21           echo "Running CI for CampusEats Task Tracker"
22           test -f README.md && echo "README found"
23
```

Problems Output Debug Console Terminal Ports Azure Postman Console Spell Checker

- PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> git switch -c chore/add-ci
- Switched to a new branch 'chore/add-ci'
- PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> git add .github/workflows/ci.yml
- PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> git commit -m "chore: add GitHub Actions CI workflow"
- [chore/add-ci f7f1608] chore: add GitHub Actions CI workflow
- 1 file changed, 22 insertions(+)
- create mode 100644 .github/workflows/ci.yml
- Enumerating objects: 6, done.
- Counting objects: 100% (6/6), done.
- Delta compression using up to 12 threads
- Compressing objects: 100% (3/3), done.
- Writing objects: 100% (5/5), 650 bytes | 650.00 KiB/s, done.
- Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
- remote: Resolving deltas: 100% (1/1), completed with 1 local object.
- remote: Create a pull request for 'chore/add-ci' on GitHub by visiting:
- remote: <https://github.com/umandathathsarani/campuseats-task-tracker/pull/new/chore/add-ci>
- remote: To <https://github.com/umandathathsarani/campuseats-task-tracker.git>
- * [new branch] chore/add-ci -> chore/add-ci
- branch 'chore/add-ci' set up to track 'origin/chore/add-ci'.
- PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> |

☰

umandathathsarani / campuseats-task-tracker

🔍 Type / to search

👤

+

🕒

🔍

📄

📧

🌐

<> Code

🕒 Issues 3

🔗 Pull requests 1

👤 Agents

🕒 Actions

📁 Projects

📖 Wiki

🛡 Security and quality

📈 Insights

⚙ Settings

chore: add GitHub Actions CI workflow #5

🟢 Able to merge

Code

🔗 Open

umandathathsarani wants to merge 1 commit into main from chore/add-ci

💬 Conversation 0

🔗 Commits 1

📄 Checks 1

📄 Files changed 1

+22

👤 umandathathsarani commented 4 minutes ago

Owner

...

No description provided.

📄 chore: add GitHub Actions CI workflow

🟢 f7f1608

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

☰

umandathathsarani / campuseats-task-tracker

🔍 Type / to search

👤

+

🕒

🔍

📄

📧

🌐

<> Code

🕒 Issues 3

🔗 Pull requests 1

👤 Agents

🕒 Actions

📁 Projects

📖 Wiki

🛡 Security and quality

📈 Insights

⚙ Settings

Actions

New workflow

All workflows

CI

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

🔍 Filter workflow runs

🕒 Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

2 workflow runs

Workflow

Event

Status

Branch

Actor

🟢 chore: add GitHub Actions CI workflow

CI #2: Pull request #5 opened by umandathathsarani

chore/add-ci

🕒 5 minutes ago

🕒 7s

...

🟢 chore: add GitHub Actions CI workflow

CI #1: Commit f7f1608 pushed by umandathathsarani

chore/add-ci

🕒 7 minutes ago

🕒 7s

...

☰

umandathathsarani / campuseats-task-tracker

🔍 Type / to search

👤

+

🕒

🔍

📄

📧

🌐

<> Code

🕒 Issues 3

🔗 Pull requests 1

👤 Agents

🕒 Actions

📁 Projects

📖 Wiki

🛡 Security and quality

📈 Insights

⚙ Settings

chore: add GitHub Actions CI workflow #5

🟢 Ready to merge

Code

🔗 Open

umandathathsarani wants to merge 1 commit into main from chore/add-ci

💬 Conversation 0

🔗 Commits 1

📄 Checks 2

📄 Files changed 1

+22

👤 umandathathsarani commented 7 minutes ago

Owner

...

No description provided.

📄 chore: add GitHub Actions CI workflow

🟢 f7f1608

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

🟢 All checks have passed

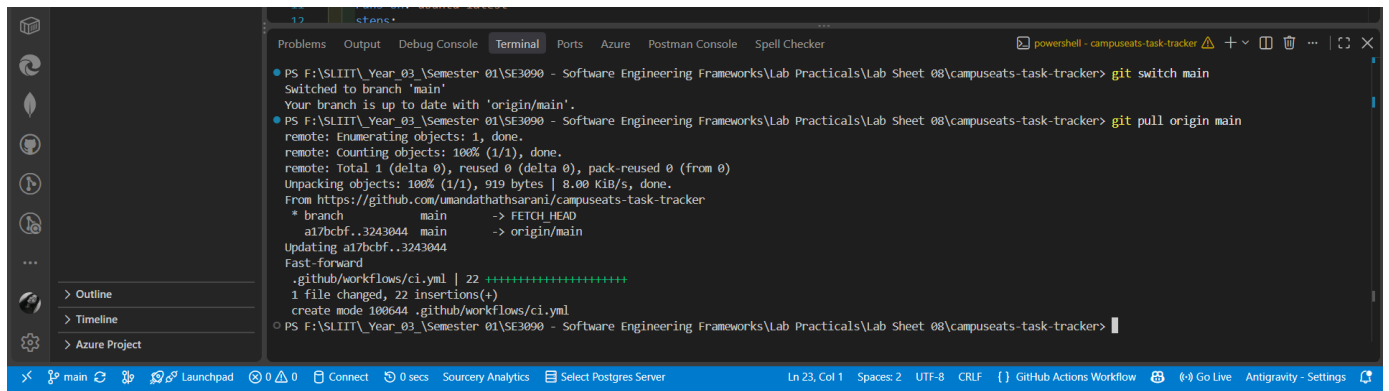
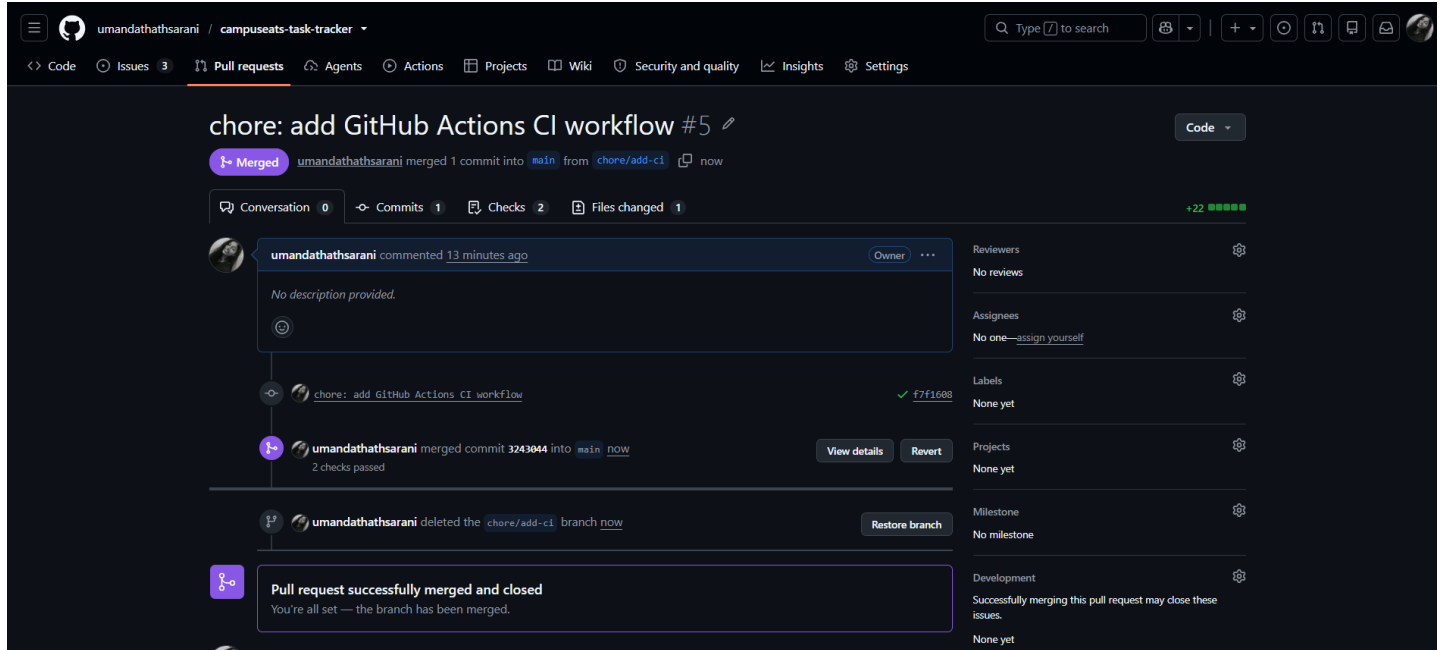
2 successful checks

🟢 No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. View command line instructions.



ci.yml

name: CI

on:

push:

branches: ["main", "feature/**", "chore/**"]

pull_request:

branches: ["main"]

jobs:

build-and-check:

runs-on: ubuntu-latest

steps:

- name: Check out the repository

uses: actions/checkout@v4

- name: List repository files

run: ls -la

- name: Basic project check

run: |
echo "Running CI for CampusEats Task Tracker"
test -f README.md && echo "README found"

Task 06 — Code quality and security review Improved

code + reflection

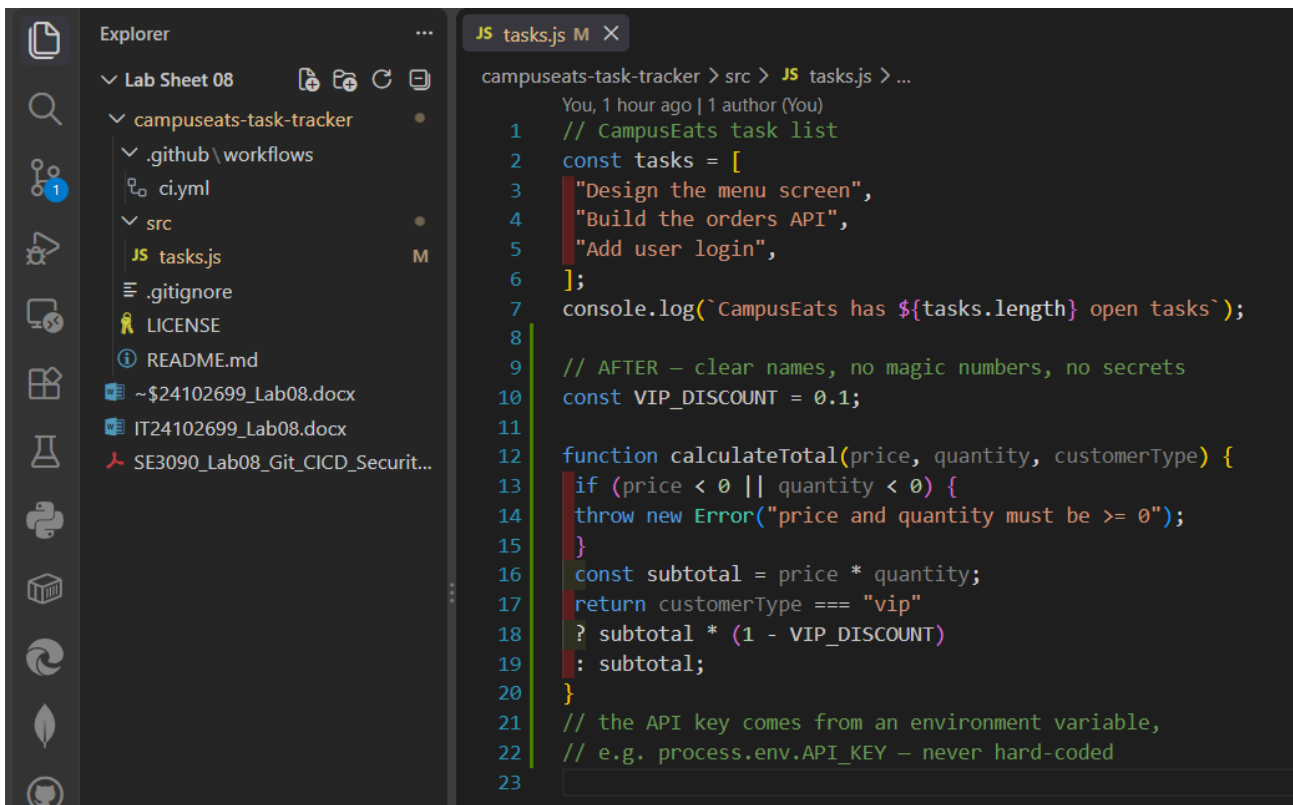
Step 1 — Review this code and list its problems

```
src/tasks.js (before)

// BEFORE — what is wrong here?
function calc(a, b, t) {
  var x = a * b;
  if (t == "vip") { x = x - x * 0.1 }
  console.log("API_KEY=sk_live_9f8a7b6c5d"); // !!
  return x
}
```

List at least **three** problems. Look for: unclear names, a “magic number”, a hardcoded secret, `var` vs `const`, loose `==`, and missing input validation.

1. **Hardcoded Secret:** The code contains a hardcoded API Key (`sk_live_9f8a7b6c5d`), which is a major security vulnerability. Secrets must never be committed to source control.
2. **Unclear Naming:** Variables like `a`, `b`, `t`, and `x`, as well as the function name `calc`, give no context about what the code actually does.
3. **Magic Numbers and Loose Equality:** The discount `0.1` is a "magic number" without context (it should be a named constant), and the code uses loose equality `==` instead of strict equality `===`. (It uses `var` instead of `const`, and it lacks input validation for negative numbers).



The image shows a Visual Studio Code editor with a file explorer on the left and a terminal at the bottom. The file explorer shows a project structure with folders like 'Lab Sheet 08', 'campuseats-task-tracker', and 'src'. The 'tasks.js' file is open in the editor, showing JavaScript code for a task list and a calculateTotal function. The terminal window shows the execution of 'git add src/tasks.js', 'git commit -m "refactor: improve code quality and remove secrets"', and 'git push origin main'. The terminal output shows the commit process, including file changes, object counting, and pushing to the remote repository.

```
campuseats-task-tracker > src > JS tasks.js > ...
You, 0 seconds ago | 1 author (You)
// CampusEats task list
const tasks = [
  "Design the menu screen",
  "Build the orders API",
  "Add user login",
];
console.log(`CampusEats has ${tasks.length} open tasks`);

// AFTER - clear names, no magic numbers, no secrets
const VIP_DISCOUNT = 0.1;

function calculateTotal(price, quantity, customerType) {
  if (price < 0 || quantity < 0) {
    throw new Error("price and quantity must be >= 0");
  }
  const subtotal = price * quantity;
  return customerType === "vip"
```

```
PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> git add src/tasks.js
PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> git commit -m "refactor: improve code quality and remove secrets"
[main cb4a26f] refactor: improve code quality and remove secrets
1 file changed, 0 insertions(+), 0 deletions(-)
PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 857 bytes | 857.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/umandathasrani/campuseats-task-tracker.git
3243044..cb4a26f main -> main
PS F:\SLIIT\Year_03\Semester 01\SE3090 - Software Engineering Frameworks\Lab Practicals\Lab Sheet 08\campuseats-task-tracker>
```

The image shows the GitHub repository page for 'campuseats-task-tracker' by user 'umandathasrani'. The 'Security and quality' tab is selected, showing an overview of security features. The left sidebar lists various security tools like Dependabot, Code scanning, and Secret scanning. The main content area shows the status of these tools: Security policy is enabled, Security advisories are enabled, Private vulnerability reporting is disabled, Dependabot alerts are enabled, Code scanning alerts need setup, and Secret scanning alerts are enabled. Each item has a link to view more details or a button to enable/disable a feature.

Security and quality

Overview

- Security policy** • Enabled
View how to securely report security vulnerabilities for this repository [View security policy](#)
- Security advisories** • Enabled
View or disclose security advisories for this repository [View security advisories](#)
- Private vulnerability reporting** • Disabled
Allow users to privately report potential security vulnerabilities [Enable vulnerability reporting](#)
- Dependabot alerts** • Enabled
Get notified when one of your dependencies has a vulnerability [View Dependabot alerts](#)
- Code scanning alerts** • Needs setup
Automatically detect common vulnerability and coding errors [Set up code scanning](#)
- Secret scanning alerts** • Enabled
Get notified when a secret is pushed to this repository [View detected secrets](#)

Dependency Check Note: I enabled Dependabot alerts in the GitHub Security tab. Since this repository has no external dependencies, Dependabot is active but currently reports zero vulnerabilities.

Step 4: Reflection

During this lab, I learned that secrets like API keys or database passwords should never be committed to source control, as doing so compromises them permanently. Instead, they should be stored securely in environment variables. I also saw how regular code review and automated CI pipelines act as critical safety nets to catch poor coding practices, magic numbers, and missing input validations before they reach the main branch. Finally, enabling automated dependency checks (like Dependabot) is essential to detect and fix vulnerable open-source packages.

Quick test — Knowledge Check

Multiple choice:

Q1	c
Q2	a
Q3	d
Q4	b
Q5	c

Git command-based:

Q6: Write the two Git commands to stage all changes and commit them with the message fix: correct total.

```
git add .  
git commit -m "fix: correct total"
```

CI/CD:

Q7: Name any three stages of a typical CI/CD pipeline, in order.

Commit → Build → Test (or Build → Test → Deploy)

Security scenario:

Q8: A teammate pushes a commit that contains a live API key in appsettings.json. What is the risk, and what three actions should they take?

Risk: The API key is fully compromised, meaning attackers or bots could find it and use it to steal data, cause damage, or incur massive financial charges.

Three actions:

1. Immediately **revoke (rotate)** the compromised API key at the provider so it no longer works.
2. Remove the key from the code and move it to a secure environment variable or secrets manager.
3. Commit the cleaned code (since it is extremely difficult to completely wipe a secret from Git history, revoking the key in step 1 is the most critical action).

